

Product, Quotient, and Power Rules for Logarithms

Answer Key

Algebra 2

Quick Answers

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|------------------------|---|
| 1. 5 | 8. $2\log(x) + 3\log(y)$ |
| 2. 3 | 9. (a) expanded form = $3\log(x) - \log(y)$, (b) |
| 3. 3 | its value = 4.7 |
| 4. $\log(5) + \log(x)$ | 10. $\log(x^3\sqrt{y})$ |
| 5. $\log(x) - \log(7)$ | 11. (a) the exact value = 6, (b) the |
| 6. $5\log(x)$ | approximate value = 1.37 |
| 7. 2 | 12. (a) the exact value = 2, — |
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What is verified

14 of 15 answers machine-checked against SymPy, across 12 problems.
 — marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSF-BF.B.5 – *problems 1, 2, 3, 4, 5, 6, 8, 9, 10, 12*

HSF-LE.A.4 – *problems 7, 11, 12*

Difficulty 1–5 of 5 across 15 tagged checks.

1. $\log_2 8 + \log_2 4$ as one logarithm, then its exact value.

Solution: Same base, so the product rule applies right to left: $\log_2 8 + \log_2 4 = \log_2(8 \cdot 4) = \log_2 32$. Then ask which power of 2 gives 32: $2^5 = 32$.

$\log_2 32 = 5$

2. $\log_3 54 - \log_3 2$ as one logarithm, then its exact value.

Solution: A difference of logarithms with the same base condenses to the logarithm of the quotient: $\log_3 \frac{54}{2} = \log_3 27$, and $3^3 = 27$.

$\log_3 27 = 3$

3. Exact value of $2\log_4 8$.

Solution: The power rule moves the coefficient inside: $2 \log_4 8 = \log_4 8^2 = \log_4 64$. Since $4^3 = 64$, the value is 3. (Equivalently $\log_4 8 = \frac{3}{2}$, and twice that is 3.) 3

4. Expand $\log_b(5x)$.

Solution: The input is a product of two factors, so the product rule splits it into a sum. $\log_b 5 + \log_b x$

5. Expand $\log_b\left(\frac{x}{7}\right)$.

Solution: The input is a quotient, so the quotient rule splits it into a difference — numerator first. $\log_b x - \log_b 7$

6. Expand $\log_b(x^5)$.

Solution: An exponent inside a logarithm comes out in front, by the power rule.

$5 \log_b x$

7. $\log_2 3 + \log_2 8 - \log_2 6$ as one logarithm, then its exact value.

Solution: Condense left to right, product first and then quotient:

$$\log_2 3 + \log_2 8 - \log_2 6 = \log_2 \frac{3 \cdot 8}{6} = \log_2 \frac{24}{6} = \log_2 4.$$

Since $2^2 = 4$,

$\log_2 4 = 2$

8. Expand $\log_b(x^2 y^3)$ completely.

Solution: Product rule first, then the power rule on each piece:

$$\log_b(x^2 y^3) = \log_b(x^2) + \log_b(y^3).$$

$2 \log_b x + 3 \log_b y$

9. $\log_b x = 1.7$, $\log_b y = 0.4$. (a) Expand $\log_b\left(\frac{x^3}{y}\right)$. (b) Its value.

Solution (a): Quotient rule, then power rule on the numerator:

$$\log_b\left(\frac{x^3}{y}\right) = \log_b(x^3) - \log_b y.$$

$3 \log_b x - \log_b y$

Solution (b): Substitute the given values into the expanded form: $3(1.7) - 0.4 = 5.1 - 0.4$. 4.7

10. Condense $3 \log_b x + \frac{1}{2} \log_b y$.

Solution: Move each coefficient inside as an exponent, then combine the sum into one logarithm of a product:

$$3 \log_b x + \frac{1}{2} \log_b y = \log_b(x^3) + \log_b(y^{1/2}).$$

$\log_b(x^3 \sqrt{y})$

11. (a) Exact $\log_5(125^2)$. (b) $\log_5 9$ from $\log_5 3 \approx 0.6826$.

Solution (a): Power rule: $\log_5(125^2) = 2 \log_5 125$, and $\log_5 125 = 3$ because $5^3 = 125$. So the value is 2×3 . 6

Solution (b): $9 = 3^2$, so $\log_5 9 = 2 \log_5 3 \approx 2(0.6826) = 1.3652$, which rounds to

1.37

12. Challenge: $\frac{1}{2} \log_2 36 + \log_2 10 - \log_2 15$. (a) One logarithm and its exact value. (b) Why the $\frac{1}{2}$ becomes a square root.

Solution (a): Move the coefficient inside first, then condense:

$$\begin{aligned} \frac{1}{2} \log_2 36 &= \log_2 36^{1/2} = \log_2 6 \\ \log_2 6 + \log_2 10 - \log_2 15 &= \log_2 \frac{6 \cdot 10}{15} = \log_2 \frac{60}{15} = \log_2 4. \end{aligned}$$

Since $2^2 = 4$,

$\log_2 4 = 2$

Solution (b): *Instructor-judged.* The power rule $\log_b(M^p) = p \log_b M$ is an identity, so it may be read right to left: a coefficient in front of a logarithm is an exponent on its input. Here p is $\frac{1}{2}$, and raising to the one-half power *is* taking the square root, so $\frac{1}{2} \log_2 36 = \log_2 36^{1/2} = \log_2 \sqrt{36} = \log_2 6$. Full credit names the power rule and applies it in that direction, rather than asserting the two expressions are equal.